GREEN SCHOOLS NATIONAL NETWORK

Climate Literacy Education Assessment Report

Summer, 2024



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Acknowledgments

The Green Schools National Network (GSNN), a national non-profit organization working with districts and schools, offers holistic support and guidance for health, equity, and sustainability. GSNN uses its GreenPrint™ (Built upon four main impact systems: Leadership; Curriculum & Instruction, Culture & Climate; and Facilities & Operations) as a frame for educational excellence[1].

GSNN partners with the following schools and districts to further health, equity and sustainability across impact systems:

- ◊ Boise Public Schools
- ◊ Boulder Valley School District
- ◊ Brooklyn Urban Garden School
- ◊ Carson City School District
- ◊ Common Ground Urban Farm School
- ◊ Encinitas Union School District
- ◊ Fayette Public School District
- ◊ Fayetteville Public Schools
- ◊ Laguna Beach Public Schools
- ◊ Los Angeles County Office of Education
- ◊ Laguna Beach Public Schools
- ◊ Mundo Verde Schools
- ◊ Oak Park Union School District
- ◊ Pilgrim School
- ◊ Portland Public Schools
- ◊ Prairie Crossing Community School
- ◊ School of Environmental Studies
- ◊ United Nations International School
- ◊ United World Colleges of Costa Rica
- ◊ Vista Grade High School
- ◊ Wayside Schools

GSNN and Portland Public Schools (PPS) have a strong mission alignment and collaborate on various fronts. This project has focused on supporting PPS to go further, farther, and faster with Climate Justice instruction and implementation. Aspects of this project have included interviewing stakeholders, meeting with PPS Climate Justice leaders, visiting school sites, participating in climate and sustainability-related events, and reviewing key documents and materials related to the pioneering work of PPS within the Climate Justice field. This scope was made possible due to a 2019 grant from Portland General Electric (PGE), the scope of which seeks to support PPS developing curriculum and resources to expand climate literacy in Portland and beyond.

[1] For a list of the 20 Core Practices of the GreenPrint™, please see Appendix.

The PPS Climate Crisis Response, Climate Justice, and Sustainable Practices Policy has three pillars. This report focuses on Pillar 3, which is to "Provide Effective Environmental and Sustainability Education," and is broken down into three sub-goals:

- ◊ Goal 3.1: PPS will empower staff as allies for a healthy climate.
- ◊ Goal 3.2: With guidance from frontline students and communities, PPS will
- ◊ develop curricular learning opportunities so PPS graduates know the causes and consequences of climate change, understand climate justice[2], and have opportunities to practice climate solutions.
- ◊ Goal 3.3: PPS staff will collaborate with students to create opportunities to
- ◊ engage youth in hands-on climate learning, preparation, and practice on a regular basis at all PPS schools.

During our interviews, visits, and the writing of this report, we strived to use these goals as a lens for our insights and opportunities.

The following reflections offer data and insight into GSNN experiences while considering the hopes and intentions of PPS leaders regarding Climate Justice integration. Content included in this document highlights key themes, strengths and assets, and insights and opportunities. In addition, notable quotes from individuals consulted were included. Lastly, GreenPrint™ core practices (of which 20) were offered as connecting points to ground the recommendations to PPS in industry best practices and support for this important work.

The GSNN overview process took place between February - July 2024 under the leadership of Kat Davis, PPS - Advisor for Climate Justice.

Associated GSNN team members involved include:

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[2] Climate justice refers to the idea that addressing climate change should be done in a way that recognizes and rectifies the unequal impacts on vulnerable communities. It emphasizes fairness, ensuring that those who contribute least to climate change, often marginalized groups, are not disproportionately affected and that solutions promote equity, human rights, and sustainability for all people. Subgoal 3.2.3 in the Climate Crisis Response Policy addresses this directly

Quotes from PPS Students

"I joined the green team because I feel like it was a good way to get younger people involved with climate change a little more".

- Maximus, 7th Grade

"I think that, from a young age, being taught that you can actually do something to combat climate change is invaluable. It's what helps people become global citizens."

- J.J., Senior

"Because everyone has a stake in stopping the climate crisis even if they don't know it yet. That's the reason that we are organizing in local schools, and not just at our federal level."

- Adah Crandall, Grant High School Graduate, NPR

"I think other people should take it seriously because even if they think it doesn't affect them, it affects everyone."

- Marshall, 7th Grade

"I feel like a lot of the lessons that they teach us don't really empower us to make change. It's kind of just like lightly touching the surface on climate change more on the effects, but not on what we can do."

- Amanda, 11th Grade

"When you learn about climate change, it can be really scary and lead to anxiety... but showing that people have made change can inspire us to create our own solutions instead of having a sense of doom. We shouldn't be scared of it and want to avoid it. We should work together, both with our peers and with people older than us, to solve things together, which can also help with the anxiety."

- Lulu, 11th Grade

"I have some family in parts where the hurricanes have been hitting. It's really scary and a wakeup call for how many that we've had... we need to have some action."

- Riley, 11th Grade

"We're the future; we're going to be the adults. Having even just a unit that really talks about climate change and climate justice would be really beneficial, especially for young people and young minds."

- Ali, 11th Grade

Executive Summary

Integrating climate justice education can enrich adopted curriculum by making it more relevant and place-based, which in turn boosts engagement, promotes equity, fosters a stronger sense of belonging, and enhances achievement in math, science, and literacy. (MacDonald, S. P. (2020), Springer (2023). Frontiers. (2020)

Portland Public Schools has passed one of the most ambitious, comprehensive, and progressive climate justice education policies in the country. It's done so in a state that has not yet mandated climate education, which creates unique opportunities and challenges. The district has developed pockets of excellence in climate knowledge and expertise, youth engagement, outdoor education, stakeholder collaboration, project-based learning, professional development, and resource development. From integrating climate justice themes across the Middle School Science Curriculum—covering energy, weather, climate, and ecology—to prioritizing youth engagement through initiatives like the Youth Climate Summit and Climate Solutions Fairs, the district has fostered environmental stewardship and action. Additionally, programs like outdoor education and the Innovation Studio Climate Justice Design Team exemplify stakeholder engagement and a systemic approach to providing a more equitable climate education. This report aims to identify ways to systematically highlight and build upon these successes, ensuring they are expanded and accessible across the district.

Portland Public Schools has the opportunity to establish itself as a national and global leader in climate justice education. This report aims to provide a strategic roadmap to help actualize that vision.

This report identifies 10 insights and 18 opportunities, all of which are interrelated and interdependent. Readers can approach this report as a menu of options, choosing where to focus resources, time, and energy to advance climate justice education in PPS. At the end of the report, we highlight and detail four high-leverage points that address multiple insights and opportunities simultaneously. These leverage points are:

1. The creation of a digital climate literacy and justice education resource hub,
2. The implementation of climate literacy and climate justice criteria for district-wide curriculum adoption,

3. The collection and visualization of climate literacy, climate justice, and sustainability data,
4. The infusion of climate literacy, climate justice education and sustainability into existing structures, initiatives, and professional development

Each of these leverage points is elaborated in detail at the conclusion of this report, offering strategic paths forward for maximizing impact.

Below is an at-a-glance summary of the insights and opportunities elaborated on throughout the report:

Insights

- ◊ #1: Balancing Curriculum Fidelity with the Need for Climate Justice Education
- ◊ #2: Limitations of Consistent Climate Education Across the District
- ◊ #3: Empowering Students Through Hands-On Sustainability and Climate Action Projects
- ◊ #4: Clarifying Language and Terms
- ◊ #5: Expanding Outdoor Education: A Pathway to Climate Justice and Community Engagement
- ◊ #6: Integrating Indigenous Knowledge for Climate Justice and Sustainability Education
- ◊ #7: Expanding Community Engagement for Broader Climate Justice Impact and Resilience
- ◊ #8: Trust Building Between District and School Building Staff
- ◊ #9: Engaging the Early Majority: Bridging the Why of Climate Justice and Education Priorities
- ◊ #10: Addressing the Need for Accessible and Curated Climate Change and Sustainability Resources in Education

Opportunities

- ◊ #1: Adoption Criteria for Climate Justice Education Integration
- ◊ #2: Leverage Existing Structures and Initiatives
- ◊ #3: Collect Data on Climate Education
- ◊ #4: Create a Climate Justice Scope and Sequence
- ◊ #5: Provide PD on the Solutionary Framework and/or 4PBL
- ◊ #6: Implement Sustainability Performance Data Collection and Visualization Programs
- ◊ #7: Clearly Define Language for Sustainability, Climate Literacy, and Climate Justice Education
- ◊ #8: Expand Access to Climate Education Professional Development
- ◊ #9: Create a Dynamic and Collaborative Digital Resource Hub to Support Climate Justice and Sustainability Education
- ◊ #10: Expand Access to Outdoor Education
- ◊ #11: Create a Digital Climate Solution Showcase
- ◊ #12: Expand the Spring 2025 PPS Climate Justice Summit
- ◊ #13: Invest in Reciprocal Connections with BIPOC, Disabled, and Minority-Led Organizations
- ◊ #14: Create and Disseminate a Climate Justice Professional Development and Messaging Toolbox
- ◊ #15: Continue to Engage Content Experts and Community Members
- ◊ #16: Gather Data on Student and Staff Mindset
- ◊ #17: Continue to Support and Expand the School Sustainability Champions Network
- ◊ #18: Provide Targeted Support and Gather Feedback from Climate Justice and Environmental Science Teachers
- ◊ #19: Create a dynamic and collaborative digital resource hub to support climate justice and sustainability education

History and Acknowledgments

In April 2016, teachers and community partners, presented a resolution to the Portland Public Schools (PPS) Board of Education's Teaching and Learning Committee calling for the integration of climate literacy into the curriculum. The resolution emphasized the need for climate education across all grade levels, promoting solutions and causes of climate change in age-appropriate ways and providing professional development for educators. It also highlighted the importance of engaging local communities, especially those most affected by climate change, and preparing students for green job opportunities.

In May 2016, the PPS Board of Education adopted Resolution No. 5272, committing to climate justice and climate change education. This resolution directed the Superintendent to work with students, teachers, and community members to develop an implementation plan to advance climate literacy, including reviewing textbooks for accuracy and eliminating those that downplayed the severity of the crisis. The Superintendent supported a teacher and community-led Climate Justice Committee to lead this process.

Educators like Tim Swinehart and Bill Bigelow, co-authors of A People's Curriculum for the Earth, played a vital role in the Climate Justice Committee, leading workshops for teachers and working toward an assessment to review and remove outdated textbooks. Community organizations, including 350PDX, were instrumental in mobilizing public support. Their collective efforts helped position PPS as a leader in climate justice and education. However, during this time frame, PPS leadership moved towards a more centralized distribution of education and resources, and by 2019 the administration chose to hire a Climate Justice Program Coordinator under the Office of Teaching and Learning to advance the Climate Literacy Resolution with central office administrators to advance curriculum adoption processes and professional development. While this choice for direct systemic integration and funding serves as a signal of the success of the community's push to advance climate literacy as a PPS priority, the trade off was a divestment of funding and support for the community-driven Climate Justice Committee that led this movement, leading to educators and community members feeling frustrated and sidelined.

In 2019, PGE provided a grant to fund open source, teacher-created climate literacy

curricula at PPS to serve as a tool for all Oregon educators. However, in 2020 the Covid-19 pandemic led to a full resource mobilization to support the basic needs and learning of students. Despite the pandemic, the Climate Justice Program Coordinator was able to successfully launch a virtual climate justice curriculum camp in the summer of 2020 to develop student and teacher co-created Climate Justice High School Elective. The elective launched in Fall 2020.

While these support for Climate Literacy Resolution continued to grow within the central administration of teaching and learning, the work taking place became less publicly visible. The withdrawal of funding and respect for the Climate Justice Committee's efforts deeply frustrated the teachers and advocates who had laid the groundwork for climate literacy in PPS. Teachers, community members, students, and parents came together to develop a draft policy, a mechanism with more accountability than a resolution, that committed PPS not only to climate literacy, but to mobilize a response to the climate crisis as a whole, centering climate justice. Students led walkouts, community partners testified at Board meetings, and the community came together to collaborate with the Board of Education and PPS staff to negotiate an ambitious climate policy.

In March 2022, the PPS Board of Education unanimously passed the Climate Crisis Response, Climate Justice, and Sustainable Practices Policy. This marked a major milestone after years of advocacy from students, educators, and activists. Key figures like Danny Cage and Jackson Weinberg rallied support for climate justice education, with students leading the charge for bold action. The policy, which aims for net-zero emissions by 2040, commits to 100% clean electricity in new buildings and retrofitting existing ones, rejecting false solutions like renewable natural gas.

The policy's passage in 2022 sparked systemic change. Superintendent Gureeroro hired an Advisor for Climate Justice positioned to report to the Chief of Staff to advise senior leadership on systems-wide strategic implementation of the Policy goals and values. Work to pilot successful climate literacy education at PPS launched with professional development opportunities for teachers that centered on teaching with a climate justice lens, Climate Justice Youth Summits seeking to connect students across the district on tangible, inspirational pathways for change, and a comprehensive assessment effort to identify strengths and opportunities for growth. While the systems-wide implementation took off with research, planning

and engagement to set up the district with a strong foundation from which to grow cohesive and impactful projects and programs to advance the policy goals, the tangible impact at the site level of this work has been slower to emerge, leading to frustration among early advocates who felt sidelined. The pilots may have feel meaningful and impactful for those in attendance, but they still only reached a small percentage of teachers and students. Thus, a parallel movement has taken place at PPS where district-wide systems are becoming more emboldened to lift policy goals and values, while certain site-based signs of success may feel harder to pinpoint, or may be felt inequitably as some sites excel at providing climate literacy learning opportunities and others lack support. This leads to a nuanced and complex perspective on the success of climate literacy education implementation at PPS, where some may feel like PPS has made significant strides, and others may critique that not enough has happened, or worse, that the district has abandoned the effort all together.



However, students remain highly engaged, with movements like the Green New Deal for Schools advocating for comprehensive climate justice education in classrooms. And teachers, whether they are inspired by opportunity or exhausted by lack of movement on climate action, continue to show up for collaborative efforts, climate justice professional development opportunities, and most of all, to support their students who are trying to navigate growing up in a climate-changed world. This report seeks to acknowledge this complex history, to recognize and honor those who have helped position PPS as a leader in climate justice education, and show the opportunities and strengths for furthering this important work both in district-wide development and in school site-level support.



Climate Justice as it Relates to PPS Top Priorities

Dr. Kimberley Armstrong has three major initiatives that are naturally aligned with the district's climate justice education efforts: Building Trust and Relationships Across our City, Advance Literacy, and Strengthen Safe, Supportive Schools[3].

The arrival of a new superintendent at Portland Public Schools (PPS) presents a timely opportunity to rebuild trust among staff, students, and community activists who have long advocated for climate justice education. The superintendent's focus on "Reestablishing Trust" can be a key step toward repairing these relationships by acknowledging past missteps and engaging those early adopters in the renewed climate literacy initiative. Involving these educators in decision-making processes, promoting transparency, and fostering open communication are crucial for restoring trust and ensuring climate justice education is meaningfully integrated across the district.

[3] Full Interview with new Portland Public Schools Superintendent Dr. Kimberlee Armstrong

[4] New Portland Public Schools Superintendent Kimberlee Armstrong shares priorities for school year, cellphones on campus and other issues

In terms of "Advance Literacy," climate literacy is a strong vehicle for fostering student engagement and improving literacy skills across reading, writing, speaking, and listening. By embedding climate-related, place-based learning into the curriculum, students gain a stronger connection to their work, which can lead to increased motivation and comprehension. The interdisciplinary nature of climate literacy offers opportunities for project-based learning, enabling students to build background knowledge and vocabulary in contextually relevant ways. As students engage with complex climate issues, they are not only developing literacy skills but also learning to think critically and become problem-solvers, preparing them for both academic and real-world challenges. This can be achieved by expanding existing teacher training opportunities and creating a digital resource and showcase hub, as outlined later in this report.

The "Strengthen Safe, Supportive tSchools" initiative aligns with the goals of climate justice education. PPS can continue to transform its schools into living laboratories where students and teachers collaborate on projects that enhance the physical environment. By involving students in efforts to improve air quality, reduce waste, and create greener, more sustainable spaces, they are empowered to contribute to tangible changes in their communities. These efforts not only align with the district's Climate Crisis Response Policy, which addresses the impacts of climate change on health and wellness, but also offer students hands-on opportunities to practice the climate solutions they learn about in the classroom. These initiatives will foster healthier learning environments and contribute to the overall physical and mental well-being of students and staff. By implementing a sustainability data dashboard more accessible for each school, the district can make sustainability performance more visible and accessible to both teachers and students, creating opportunities for them to become active changemakers in their schools. This can lead to positive mental health benefits, as well as improvements in physical well-being and safety.

At the heart of these three initiatives—building trust, advancing literacy, and strengthening schools—is the mental and physical well-being of students and staff, especially in the face of eco-anxiety. Eco-anxiety, which many young people experience as they grapple with fears about the future of the planet, makes it difficult for them to envision a hopeful and achievable future. As educators, it is our responsibility to foster what some experts call critical hope: the ability to acknowledge the challenging realities of our current climate crisis while still empowering students to work towards a positive future. The work of climate literacy

directly addresses this by offering students opportunities to collaborate on local, place-based solutions, allowing them to take tangible actions in their communities.

Through project-based learning, outdoor education, and solutionary thinking, students not only develop essential academic and problem-solving skills but also cultivate a sense of efficacy and agency. This pairing of strong curriculum with the greening of physical spaces creates resilient schools that are better equipped to face climate disruptions and extreme weather, while simultaneously supporting the physical and mental well-being of students. The fact that all of this is grounded in PPS's Climate Crisis Response Policy is a significant step forward, and the following sections of this report will outline the insights and opportunities to fully implement this policy and its supporting initiatives.



Strengths, Assets and Successes

First Annual PPS Innovation Summit

The First Annual PPS Innovation Summit gathered students, staff, board members, and community organizations to explore strategies for systemic change through the theme of Productive Disruption. Trenelle Doyle, Founder of Go Girl Ride, described this as disrupting the status quo with intention and impact, while Bryan Lee Jr. of Colloqate Design emphasized reshaping not just physical spaces but also cultural and social narratives toward justice. The summit prioritized voices with lived experiences of oppression, centering equity through panels, student speeches, and a keynote. The PPS Innovation Studio is now tasked with empowering the community to implement these creative, impactful changes.

Student Climate Action Design Team

In Spring 2024, the Student Climate Action Design Team launched, composed of students, educators, community climate organizations, and staff working to develop pathways that empower students to take meaningful climate action within Portland Public Schools (PPS). In collaboration with the PPS Innovation Studio, the team employs the Innovation Equity Design Process, integrating design thinking with racial equity and social justice. Despite challenges, especially for historically underrepresented students, the team is focused on creating accessible, diverse pathways for all students to engage in climate action, particularly those most affected by climate change.



School Sustainability Champions Network

In Fall 2023, PPS launched the School Sustainability Champions Network, a group of teachers and staff who meet monthly to discuss climate-related initiatives across the district. The Champions have led

impactful events, programs, and climate art projects at various schools, while also sharing ideas, resources, and solutions to common challenges. Serving as a vital communication link, they help spread information about climate programs, events, and opportunities for student and teacher involvement, fostering collaboration and inspiration throughout PPS.

Climate Solutions Fairs

For the last four years, PPS has partnered with SAGE (Seniors Advocates for Generational Equity) to host Climate Solutions Fairs at middle schools in PPS. In Spring 2024, SAGE hosted fairs at Roseway Heights Middle School and Harriet Tubman Middle School. These fairs provided students with the opportunity to learn about local organizations and sustainability practices aimed at mitigating climate change and building community resilience in the Portland area.

Second Annual PPS Student Climate Summit

In May 2024, more than 100 students from seven PPS high schools gathered at the Redd Building on Salmon Street for the Second Annual PPS Spring Climate Summit. The event, themed Seeds of Change, aimed to inspire student leaders and connect them with community organizations to support climate action in their schools. Students learned practical skills for managing climate projects with a focus on impact and equity.

Keynote speakers included Oregon Representative Khanh Pham, climate author Elizabeth Doerr, staff from the Portland Clean Energy Fund (PCEF), Kat Davis (PPS Advisor for Climate Justice), and student leaders from the Climate Justice Youth Advisory. The Seeds of Change theme will continue into the next school year, as PCEF begins financially supporting student-led climate action projects across the city, fostering further collaboration between students, community organizations, and district staff.

Climate and Sustainability Initiatives in PPS High Schools

High schools across Portland Public Schools (PPS) have taken significant strides in climate action and sustainability, driven by student leadership and community engagement. These types of initiatives should be shared, scaled and replicated as PPS builds capacity.

At **Grant High School**, students in the Green Club made PPS history by launching the district's first high school front-of-house **food scrap collection program** on March 4th, 2024. While PPS kitchens have long managed back-of-house food waste, creating a system for students to participate in food scrap collection in the cafeteria proved challenging. Green Club students worked diligently with administrators, secured volunteer hours for monitoring, ordered bins, and created informational signage. Their persistence led to a successful launch after receiving staff approval, marking a milestone for sustainability initiatives in the district.

Ahead of Earth Day, **Alliance High School** hosted a guided tree walk led by Jim Gersbach, showcasing a **Learning Landscape** developed in partnership with the Concordia Tree Team and Portland Parks & Recreation Urban Forestry. This landscape features around three dozen tree species, including deciduous and evergreen oaks, as well as relict species. A highlight of the event was the chance to see a living specimen of Oregon's official state fossil, deepening participants' connection to the region's natural history.

Earth Day celebrations at various PPS high schools further exemplify the commitment to environmental stewardship. **Franklin High School** hosted a campus cleanup, while **Grant High School's Climate Justice Club** organized a cornhole tournament and distributed Earth Day buttons to raise awareness. **McDaniel High School's Naturalists Club** participated in the Roseway Neighborhood Earth Day Celebration with an informative tabling event. At **Roosevelt High School,** Environmental Science and Leadership classes collaborated to create an Earth Day carnival, where students engaged in climate change activities and signed a climate pledge in exchange for frozen yogurt.



The **Climate Justice Youth Advisory**

(CJYA), composed of passionate high school students from across PPS, plays a critical role in shaping the district's climate action agenda. CJYA members were instrumental in organizing the Spring Climate Summit, providing input on event themes, speakers, and tabling organizations, thereby amplifying student voices in climate justice work.

In February 2024, several high school students from **Benson, Franklin, Grant**, and **McDaniel** traveled to Salem for a day of **legislative advocacy**. They participated in a floor session at the House of Representatives, learned about the legislative process, and met with state representatives to advocate for public education and youth empowerment, including bills supporting summer learning programs, youth advisory councils, and expanded school-based health care services.

Middle School Front-of-House Food Scrap Programs

Front-of-house (FOH) food scrap programs in PPS middle schools have laid a strong foundation for sustainability efforts across the district. Beverly Cleary Middle School has set a notable example, where students actively participate in FOH composting by assisting custodians in sorting food scraps. This hands-on involvement has fostered environmental stewardship and built awareness among students. While Grant High School recently became the district's first high school to implement a similar program, several elementary and middle schools had already embraced FOH composting prior to 2020.



Climate and Sustainability Initiatives in PPS Elementary Schools

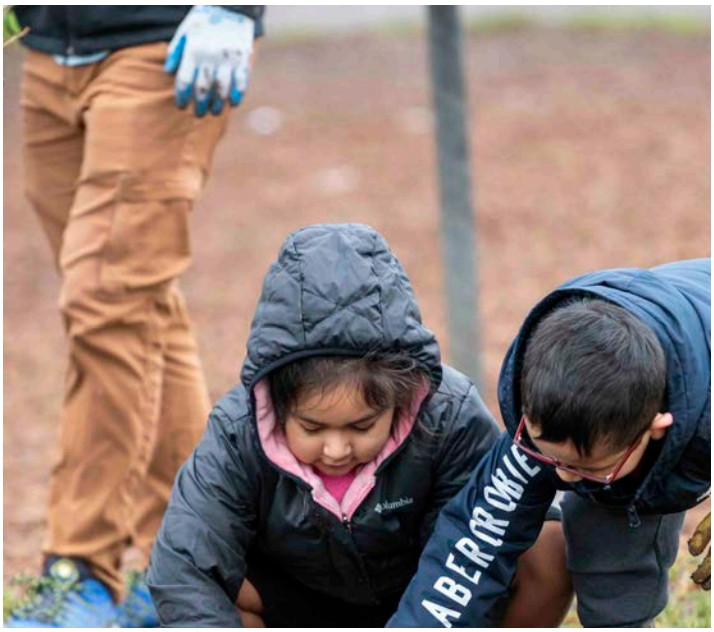
As a result of key community partnerships, elementary schools in Portland Public Schools (PPS) are leading the way in climate action through hands-on projects, community engagement, and sustainability-focused events.

Eco-School Network coordinates volunteers at elementary schools

across Portland, engaging thousands of students in hands-on learning about gardens, alternative transportation, waste diversion, environmental stewardship and more.

At **Rose City Park Elementary**, a partnership with Grow Portland, Full Cellar Farm, and Wild Roots Farm launched a new **food waste sorting program**. This initiative kicked off with a fun-filled day where students and community members taste-tested root vegetables, promoting sustainable food practices and reducing waste.

Peninsula Elementary hosted a large-scale **Earth Day event** that involved over 150 students, families, and volunteers. Participants cleaned up the neighborhood, moved soil, planted an edible garden, and removed invasive species. The effort resulted in 1,600 pounds of litter and yard waste being cleared from the area, contributing to the school's environmental and outdoor learning spaces. This project was supported by community partners Depave and SOLVE, highlighting the power of collaboration.



At **Buckman Elementary**, Earth Day was celebrated with an **Earth-Day cleanup** event followed by a silent disco, co-hosted by SOLVE and Heartbeat Silent Disco, combining environmental stewardship with fun, community-building activities.

Bridger Creative Science School marked Earth Day 2024 with a Sunday afternoon of **community stewardship**, where families participated in clothing swaps, made upcycled planters and bird feeders, and spread mulch in the school's Rain Garden. The event also served as an opportunity to envision a revamped garden education space and additional outdoor learning areas.

At **Woodstock Elementary**, the school community celebrated the installation of a **Peace Pole and Peace Garden** with a special Earth Day ceremony. The Peace Pole, which displays the message "May Peace Prevail on Earth" in multiple

languages, symbolizes the school's commitment to peace, sustainability, and global understanding. This initiative, led by Principal Johnson and Mr. Hunter's class, adds a meaningful cultural and environmental element to the school garden.

Fourth-grade students from **Scott Elementary** recently visited two local farms to explore unique food production practices. At **Live Local Organic**, they learned about sustainable farming through a combination of aquaculture and hydroponics. At **Black Futures Farm**, a community-focused organization led by Black-identified and African Diasporic individuals, students experienced how food production fosters social connections and cultural resilience. This field trip highlighted the importance of sustainability, community, and local food systems in the students' education.

Curriculum Integration and Pedagogy: Insights

Insight #1: Balancing Curriculum Fidelity with the Need for Climate Justice Education



GreenPrint Core Practice #6: Schools plan strategically for curriculum and instruction that promote health, equity, and sustainability

Because the PPS commitment to a climate justice curriculum is ahead of the curve for school districts, the curriculum that the district adopts from nation-wide sources may not make the connections about the inequitable human-environmental impact of climate change, or offer resources for teaching the climate-related curricula that attend to students' mental health that the district's policy demands. There is a noticeable tension between adhering strictly to the adopted curriculum in most subject areas and the desire to integrate climate justice education. This tension is felt even in science classes, where climate change is explicitly included in the standards, but questions remain about whether, how, and to what extent climate justice should be taught. In interviews, Many educators expressed feeling constrained by standardized content requirements and fear deviating from the prescribed curriculum. Educators sometimes incorporate climate justice "under the radar" for fear of district reprimand or retaliation. This may limit opportunities to engage students in deeper discussions about the systemic causes of climate change, environmental racism, and social inequities. At the same time, there is a growing recognition that climate justice education is essential to preparing students for the complex environmental and social challenges they will face. To address this, there is a need for clearer guidance and resources on how to weave climate justice seamlessly into existing curricula without compromising fidelity, empowering teachers to explore these critical issues confidently and effectively..

Insight #2: Limitations of C onsistent Climate Education Across the District

GreenPrint Core Practice #10: Curriculum and instruction leaders build a comprehensive and coherent system of assessment that measures sustainability literacy

Based on student anecdotes, when climate change is taught, it is typically brief and limited to the basics—causes and effects—with little discussion of solutions, justice, or actionable steps. Most said that any mention of climate change was in science class, and a few students said there wasn't much difference in the content of climate change instruction, even at different grade levels. According to some students, this approach sometimes left them feeling disempowered and uninspired, missing the opportunity to engage them in meaningful discussions about their role in addressing climate challenges and advocating for equity.

Insight #3: Empowering Students Through Hands-On Sustainability Projects



GreenPrint Core Practice #9: Schools use instructional practices that include phenomena, place, project, and problem-based learning

The facilities department at Portland Public Schools (PPS) has a small, dedicated sustainability team making significant strides in reducing the ecological footprint of school buildings despite being stretched thin by their heavy workload of maintenance and retrofitting. While they currently lack the capacity and expertise to engage directly with students and teachers, they can bridge this gap by involving students in sustainability efforts. With the guidance of the School Sustainability Champions Network, facilitated by the Climate Resilience Program Manager, Ari

Ettinger, community partnership organizations, students can actively participate in initiatives to lower the ecological footprint of their buildings and campuses. This approach becomes especially powerful when sustainability data—such as energy use, water conservation, and waste reduction metrics—is accessible and transparent. By turning these authentic examples into classroom projects, students can learn hands-on how to analyze and act on sustainability challenges, creating a more integrated and impactful educational experience .

Insight #4: Clarifying Language and Terms

GreenPrint Core Practice #7: Schools create the conditions for leaders, teachers, and students to plan and implement high-quality curriculum



During our visit, we noticed some confusion and even tension in the language used to describe the goals of the climate crisis response policy. There seemed to be a discrepancy between the terms 'sustainability,' 'climate change,' 'climate justice,' and 'environmental justice.' Some stakeholders used these terms interchangeably, while others strongly preferred focusing on one over the other. Sustainability often focuses on clean air, clean water, conservation, and technical solutions, emphasizing individual consumer actions. 'Climate change' typically centers on the causes and effects of global warming, often focusing primarily on the scientific understanding of these phenomena. In contrast, climate justice addresses the root causes of climate change, such as fossil fuel dependency, industrialization, capitalism, economic growth at the expense of people, colonial and historical injustices, inequality and poverty, political inaction, overconsumption, agricultural practices, and environmental racism.

‘Environmental justice’ emphasizes the fair treatment and meaningful involvement of all people in environmental policy decisions regardless of race, income, or background. It addresses the disproportionate impact of environmental hazards on marginalized communities, highlighting the need for equitable solutions that protect the most vulnerable populations from pollution, unsafe living conditions, and the effects of climate change.



In addition, there is some confusion with the phrase “School Climate,” which is about a school’s general culture and atmosphere but has little to do with the four terms above.

Insight #5: Expanding Outdoor Education: A Pathway to Climate Justice and Community Engagement

GreenPrint Core Practice #12: Schools promote healthy relationships between peers, adults, the local community, and the natural world

Interviews and observations have revealed a noticeable pattern of limited outdoor education opportunities across the district, with only a few elementary schools offering such experiences. For instance, schools with dedicated gardens, particularly Sunnyside Elementary, stand out as exceptions where outdoor education is thriving. At Sunnyside, students engage in garden activities and walk field trips to gather data and work on projects that enhance their neighborhood beyond the school grounds. Replicating this model across the district could significantly benefit students and teachers, fostering hands-on learning, strengthening community connections, improving mental health, and advancing climate justice and sustainability goals.

Expanding these opportunities would empower more students to experience and understand their local environments, encouraging a deeper commitment to sustainable practices and environmental stewardship. Some schools with gardens have limited capacity because the upkeep often depends on parent volunteers. Notably, Sunnyside Elementary has a full-time garden coordinator, funded through PTA-raised money.

Insight#6: Integrating Indigenous Knowledge for Climate Justice and Sustainability Education

GreenPrint Core Practice #8: Schools use an integrated approach to curriculum design that reflects the interdisciplinary nature of understanding the world around us.

Indigenous Traditional Ecological Knowledge (ITEK) refers to the cumulative, holistic understanding of the environment developed by Indigenous peoples over generations. Rooted in deep relationships with the land, water, plants, and animals, ITEK encompasses practices, beliefs, and wisdom passed down through oral traditions, ceremonies, and lived experiences. It emphasizes balance, reciprocity, and stewardship of the natural world. ITEK is vital to climate justice as it highlights the importance of Indigenous leadership in addressing environmental challenges. Indigenous communities, often on the frontlines of climate change, use their knowledge to protect ecosystems and advocate for sustainable practices that counter the environmental harm caused by industrialization, colonialism, and extractive economies. By embracing ITEK, we move away from a narrative of separation between humans and nature, toward one of interconnection and reciprocity, where humans are seen as integral to, rather than separate from, the natural world. This shift is essential for creating equitable, sustainable climate solutions.

Oregon State partnered with local Indigenous tribes to co-develop the Tribal History, Shared History curriculum to honor Oregon's Indigenous communities and ensure students receive a more accurate, inclusive education. This collaboration involved working closely with tribal leaders, educators, and historians to incorporate authentic Indigenous perspectives, knowledge, and histories into the curriculum. By centering the voices of the tribes, the curriculum reflects the rich cultural heritage and ongoing contributions of Indigenous peoples in Oregon while also addressing the lasting impacts of colonization and historical injustices.

Curriculum Integration and Pedagogy: Opportunities

Opportunity #1: Adoption Criteria for Climate Justice Education Integration



GreenPrint Core Practice #6:
Schools plan strategically for curriculum and instruction that promote health, equity, and sustainability.

Portland Public Schools' curriculum adoption process consists of four key steps:

1. **Define**: In this initial phase, adoption leads develop a project plan, engage committee members, conduct an internal program evaluation, and establish the adoption scope. They also create a best practices paper and a materials evaluation matrix.
2. **Select**: In this phase, the adoption committee reviews publisher programs based on priorities set in the Define phase. The committee narrows down options using guiding questions and a standards trace to determine which programs should move forward in the adoption process.
3. **Implement**: This step focuses on rolling out the selected instructional resources to teachers and students. It includes professional development, support for instructional pedagogy, and ongoing training throughout the school year, as well as for new teachers or those changing subjects or grade levels.
4. **Evaluate**: Evaluation occurs throughout all phases, focusing on both the effectiveness of the instructional resources and whether they meet student needs. Both quantitative and qualitative data are collected to inform this ongoing assessment.

Develop and implement adoption criteria that mandate the integration of climate literacy and justice education. If the current curriculum does not meet these criteria, adopt supplemental resources and materials to fill the gaps. More details for this approach are elaborate in the Leverager Points section at the end of this report.

Opportunity #2: Leverage Existing Structures and Initiatives

GreenPrint Core Practice #7: Schools create the conditions for leaders, teachers, and students to plan and implement high-quality curriculum and instruction for health, equity, and sustainability.

Utilize current programs that align with climate justice education, such as Social Emotional Learning (SEL), Tribal History/Shared History, Career Technical Education courses and the Middle School Redesign. These initiatives lend themselves to the infusion of climate literacy.

Middle School Redesign

Portland Public Schools identified 18 initiatives across five themes—Integrated Adolescent Supports, Inclusive and Differentiated Teaching and Learning, Embracing Change, Racial Equity and Social Justice, and Professional Excellence and Support—aimed at transforming outcomes for middle-grade students. These initiatives are designed to support adolescent learners and improve the school experience for all 6th-8th graders. Each year, the district will prioritize and deeply support two to three of these initiatives.

SEL in Elementary School

Social and Emotional Learning (SEL) is a process that helps individuals, both young and old, acquire the knowledge, skills, and attitudes necessary to understand themselves, connect with others, achieve goals, and support their communities. The CASEL framework outlines five core competencies essential for SEL: self-awareness, self-management, social awareness, relationship skills, and responsible decision-making. These competencies help individuals manage emotions, empathize with others, build meaningful relationships, and make thoughtful choices. SEL development occurs throughout life and is influenced



by supportive environments, including classrooms, families, and communities. By integrating SEL into educational and social settings, we can foster inclusive, engaging, and supportive learning spaces that prepare young people to contribute positively to society.



Other schools and districts have used SEL as a framework for integrating climate literacy in a developmentally appropriate way, especially in elementary schools. Responsible decision-making, one of the core SEL competencies, naturally aligns with the principles of doing the most good and least harm for people, the planet, and animals. By encouraging students to consider the broader impact of their choices, teachers can weave in concepts of sustainability and environmental stewardship without directly mentioning terms like climate change or global warming. Relationship skills, another key SEL component, can be developed as students collaborate on projects aimed at improving their schools and communities, such as making them cleaner, safer, and greener. These hands-on efforts foster a sense of self-efficacy and confidence in young people, empowering them to take action. There are many academic and transdisciplinary connections that can be made with this project-based learning approach that can tie SEL together with academic skills and sustainability. By incorporating Traditional Ecological Knowledge (TEK) and biophilia—the innate human connection to nature—teachers can enhance SEL while fostering a love and respect for the environment. Since SEL is often a district-wide initiative, it serves as a powerful leverage point for embedding climate literacy into everyday learning, laying the groundwork for a more sustainable future.

Career Technical Education (CTE) and other Workforce Pathways

The rise of green jobs in the U.S. is reshaping the workforce, expanding beyond traditional sectors like solar and wind energy to include industries such as construction, medicine, architecture, and even finance. As companies increasingly



prioritize sustainability due to customer and shareholder demands, the need for workers with green skills has grown. In 2022, over 410,000 job postings were for "core green" roles, and it's expected that the demand for these positions will grow by 5.5% over the next five years. Beyond these core green jobs, there's a significant demand for "green-enabled" jobs in non-traditional fields that require at least one green competency. With over 51.6 million workers who could potentially be reskilled for these positions, the green economy presents enormous opportunities for workforce transformation. Upskilling programs and green job initiatives are vital to meeting the needs of this expanding sector, allowing workers to transition into roles that can drive sustainability and climate action across the economy.

Portland Public Schools (PPS) can leverage Career and Technical Education (CTE) courses and other workforce development pathways in high schools to infuse climate literacy into the curriculum in ways that connect to students' future careers and livelihoods. By reframing every job as a potential green job, educators can help students see how any interest or passion can be translated into a climate solution. This systemic integration of climate literacy can energize high school coursework and provide relevance to students' lives beyond the classroom. While some high schools already offer CTE programs related to climate and sustainability, there are vast opportunities to scale and multiply these efforts. Expanding these examples can provide students with a broader understanding of how climate solutions can permeate all industries, preparing them for a future where green skills are increasingly necessary.

Opportunity #3: Collect Data on Climate Education



GreenPrint Core Practice #10: Curriculum and instruction leaders build a comprehensive and coherent system of assessment that measures sustainability literacy and promotes health and equity.

To address the limited teaching of climate change and justice, the district could gather data from students and teachers to map out where and how these topics are currently covered in the curriculum.

Surveys, focus groups, and classroom observations could identify gaps in content, instructional practices, and areas where climate justice is effectively integrated. This data can then inform targeted improvements, such as developing specific curriculum modules, providing professional development, and creating interdisciplinary connections, ensuring a more comprehensive and impactful approach to climate education across all grade levels and subjects.

Opportunity #4: Climate Justice Scope and Sequence:

GreenPrint Core Practice #6: Schools plan strategically for curriculum and instruction that promote health, equity, and sustainability.

To effectively integrate climate change and justice into the curriculum, a comprehensive scope and sequence should be created based on the data gathered from students and teachers. This scope and sequence would map out all existing instances where sustainability and climate justice are already taught across K-12 grades, providing a clear overview of current efforts. Additionally, it would identify natural integration points within the curriculum, education that builds on students' knowledge year after year. This plan would also help design professional development for teachers and develop supplemental materials that align with district goals. This

can likely be done with existing mapping software used by the district.



Opportunity #5: Solutionary Approach (4PBL)

GreenPrint Core Practice #9: Schools use instructional practices that include phenomena-, place-, project-, and problem-based learning.

Encouraging a solutionary approach rooted in 4PBL empowers students to advocate for improving their school's sustainability performance by tackling real-world environmental challenges on their campuses. With guidance from educators and support from community partners, students can analyze accessible sustainability data, such as energy use or waste management metrics, and develop actionable plans to reduce their school's ecological footprint. This hands-on method builds critical thinking and advocacy skills and connects classroom learning with real-world impact, fostering a sense of ownership and responsibility. Students become active change-makers in creating a more sustainable school environment by engaging in these projects. The older a student is, the more reach they can have in making a change. The Portland Clean Energy Fund Climate Friendly Public Schools grant funding youth climate action is an example of a solutionary approach.

Opportunity #6: Sustainability Performance Data Collection Programs

GreenPrint Core Practice #10: Curriculum and instruction leaders build a comprehensive and coherent system of assessment that measures sustainability literacy and promotes health and equity.

Use programs like START to make sustainability metrics, such as energy use, waste production, and water conservation, visible and accessible to students and teachers, paired with tailored educational materials that explain these metrics in context.

Making this data transparent communicates the urgency of sustainability and climate justice efforts. It tracks progress, engaging key stakeholders like families, board members, and the broader community in the school's environmental initiatives. Tracking student-led sustainability projects through START can help build a school-wide culture of environmental responsibility, where students see the direct impact of their actions. Additionally, integrating certification programs like EcoSchools U.S., which supports environmental learning and climate action from elementary through secondary levels, can provide a structured framework for schools to advance their sustainability goals and recognize achievements. These combined efforts can transform schools into active hubs for climate action, fostering a community dedicated to sustainability.

Opportunity #7: Clearly define language

GreenPrint Core Practice #5: Leaders create and maintain a communication system that engages the whole community in support of healthy, equitable, and sustainable schools.



Create a set of foundational documents that outline a clear progression of definitions, from sustainability on one end to climate and environmental justice on the other. These documents would illustrate how different approaches, from individual actions focused on environmental conservation to systemic changes that address the root causes of climate change, all contribute to a greener future. Additionally, these resources should communicate that there is a place for everyone in this continuum, encouraging inclusive participation from all students and teachers.

A helpful starting point might be the newly released NOAA Climate Literacy Guide, updated from 2009, that now incorporates climate justice and a lens of equity.

Opportunity #8: Expand access to climate education professional development



GreenPrint Core Practice #7: Schools create the conditions for leaders, teachers, and students to plan and implement high-quality curriculum and instruction for health, equity, and sustainability.

To support this effort, the district could continue to provide targeted professional development, workshops, and resources that help educators understand what is developmentally appropriate at different grade levels and how to integrate these concepts into their teaching. The district can create a more unified and effective approach to climate education by offering clear definitions and supportive materials.

Example: June 2024 PPS Climate Justice Professional Development Day

This professional development opportunity introduced the Climate Crisis Response Policy and featured a visit from David DenHartog of GSNN, who connected these efforts to a broader national movement. Participants also received instruction from Tim Swinehart, a teacher at Lincoln High School and one of the district's earliest advocates for climate justice education. For many years, Tim felt let down by the district, as they were either unable or unwilling to incorporate the climate education he knew was urgently needed. However, with a few years of Policy implementation, the district could finally highlight Tim's expertise, creating a moment of repair. Tim led the group in a thoughtful reflection on their experiences with the Willamette River, followed by a gallery walk illustrating the river's history of environmental injustice. It was clear that teachers benefit from learning from peers trained in this vital field of education whenever possible.

The field trip with the Braided River Campaign continued along the Willamette River, where participants explored issues like dredging, pollution, displacement of riverside communities, the history of recreation on the river, and the city's prioritization of industry over people. Though the topics were weighty, the trip concluded at the Braided River Campaign's vibrant and creative space in the Lloyd Center Mall, designed to inspire ongoing education, advocacy, and reflection.

Opportunity #9: Expand Access to Outdoor Education



GreenPrint Core Practice #12: Schools promote healthy relationships between peers, adults, the local community, and the natural world.

Integrating outdoor education into the curriculum allows students to engage directly with real-world sustainability issues, fostering a deeper understanding of the environment and their role in protecting it. By moving beyond the classroom, students can explore local ecosystems, observe the impacts of climate change, and participate in hands-on activities like gardening, composting, or water quality testing.

These experiences make abstract concepts more concrete and meaningful, encouraging students to think critically about human-environment interactions and sustainability practices. Outdoor education also promotes experiential learning, collaboration, and problem-solving, empowering students to become active stewards of their communities and the planet.

Examples of Outdoor Education in PPS:

Portland Public Schools (PPS) has a rich landscape of outdoor education, with numerous schools incorporating hands-on learning experiences in gardens and natural spaces. **Lent Elementary** stands out with its long-term partnership with Growing Gardens and a vibrant garden space that has flourished for over a decade. Their Garden Educator created the **Semillas Indigenas** (Indigenous Seeds) curriculum, which brings cultural and environmental learning together. **Scott Elementary** also partners with Growing Gardens, connecting their school garden to the cafeteria, ensuring students see the full cycle of growing and consuming food. **Harrison Park** collaborates with Grow Portland, incorporating culturally affirming foods into their garden beds to enrich student learning. At the high school level, **Alliance HS** offers a Natural Resources class, and **McDaniel HS** runs a **CTE Sustainable Agriculture** program, providing students with in-depth, practical experiences in sustainability. Across PPS, 16 schools partner with Growing Gardens or Grow Portland, and an estimated ten additional schools offer outdoor lessons through PTA volunteers, with many more participating in garden clubs through SUN Schools. These programs collectively strengthen student connections to nature and sustainable practices. Their stories should be amplified and scaled across the district.



Community and Stakeholder Engagement: Insights

Insight#7: Expanding Community Engagement for Broader Climate Justice Impact and Resilience

GreenPrint Core Practice #11: Schools plan strategically for a school culture and climate that promotes health, equity, and sustainability mindsets

There are many examples of engaging various stakeholders and community partners, as evident in the Youth Climate Justice Summit, the Climate Solutions Fair, and the Climate Justice Design Team. These initiatives highlight the work of innovators and early adopters. Developing a framework that expands these initiatives to reach a broader audience is essential to scaling up these efforts.

Insight #8: Trust Building Between District and School Building Staff

GreenPrint Core Practice #5: Leaders create and maintain a communication system that engages the whole community in support of healthy, equitable, and sustainable schools.



Based on conversations with a handful of aggrieved teachers, there appears to be a significant lack of trust between Portland Public Schools (PPS) district administration and some of its staff, particularly among early adopters and activists who have worked closely with local community members, students, and families to advance climate justice initiatives. These educators feel let down by what they perceive as a lack of transparency and the mismanagement of funds, often directing blame toward specific district staff members. Some teachers expressed frustration about

inadequate communication around events like the climate justice summit, which they felt was not inclusive. There have also been instances of confusion over accountability, where teachers mistakenly blamed current staff for decisions made by former employees. Additionally, some teachers reported feeling reprimanded by the district for implementing climate justice policies that did not align with the district's vision, leading to a sense that this work had to be done in secrecy or not at all. However, it is important to note that these impressions are based on a few teachers, and it is unclear how widespread this distrust is. Moreover, the landscape has become more favorable since the adoption of the CCRP. We observed that district staff are acutely aware of these concerns and are actively taking steps to address them and repair relationships.

Insight #9: Engaging the Early Majority: Bridging the Why of Climate Justice and Education Priorities



GreenPrint Core Practice #1: Schools practice visionary leadership.

There is a growing need for more targeted messaging and professional development around the importance of climate justice and sustainability in education. While a core group of dedicated, knowledgeable, and passionate innovators and early adopters are already committed to this work, engaging the early majority is the challenge. To bridge this gap, it's essential to connect climate justice and sustainability to the priorities and responsibilities of teachers and educational leaders. By aligning these critical issues with what educators care about—such as student well-being, equity, and preparing future-ready citizens—we can create a broader understanding and commitment to integrating sustainable practices into schools and curricula. Through strategic communication and professional development, we can foster a culture that values and prioritizes climate justice as an essential education component.

Community and Stakeholder Engagement: Opportunities

Opportunity #10: Digital climate solution showcase



GreenPrint Core Practice #5: Leaders create and maintain a communication system that engages the whole community in support of healthy, equitable, and sustainable schools.

Create a centralized online platform that highlights the work of innovators and early adopters of climate justice education at PPS. This showcase would feature videos, project highlights, interactive presentations, and stories from various stakeholders, including students, educators, and community partners, demonstrating their contributions to climate justice efforts. By offering an engaging, visually rich experience and providing resources and tools for replication, the digital showcase would help expand these initiatives, reaching a broader audience and inspiring action across the district and beyond. See below in the section titled "Resource Development and Accessibility" for elaboration on digital resources.

Opportunity #11: Expand the Spring 2025 PPS Climate Justice Summit

GreenPrint Core Practice #3: Schools promote a culture of community and belonging.

Teachers have expressed a strong desire to bring more students to the Spring Climate Summit in recent years than the venues can accommodate. Last year, each high school was allotted 20 spots, but many teachers wanted to bring their entire classes, which often numbered around 145 students. Limiting the number of attendees prevents broader student engagement with climate justice—an issue deeply relevant to their lives. To make the summit more accessible and inclusive, inviting as

many students as possible is essential. We aim to make this event an entry point for students who may not typically be drawn to the topic and ensure that it aligns with how schools operate, recognizing that it's easier for teachers if all their students can participate.

The district could consider three approaches to expand access to the PCEF Spring 2025 Climate Summit:

- First, schools could be supported in organizing their climate justice summits, creating multiple concurrent events across different locations. This decentralized model would engage more students, educators, and community members directly at their sites, fostering local ownership and relevance.
- Second, the district could envision a larger, centralized event in a more spacious venue, allowing for more attendees and inviting a wider range of stakeholders.
- Lastly, a virtual attendance option could be offered to accommodate those unable to attend in person. While in-person participation is preferable, especially for climate justice advocacy, as it fosters stronger community bonds and deeper engagement, a virtual component would ensure no one is excluded due to logistical or capacity constraints. Both the decentralized and larger centralized events, combined with a virtual attendance option, would increase access and equity, ensuring more voices are heard and empowering more participants to contribute to climate action.

Opportunity #12: Invest in reciprocal connections with BIPIOC, disabled, and minority-led organizations and students who identify within these groups



GreenPrint Core Practice #4: Schools cultivate strong school-family-community partnerships.

To understand better how to make climate justice education equitable, the district needs to follow through when



using words like "justice" and "equity" by deeply engaging marginalized students and communities to understand their needs and how to create a climate justice culture that is inclusive, culturally resonant, and diverse. Special attention must be paid to creating reciprocal rather than extractive or performative relationships.

Opportunities #13: Professional Development and Messaging Toolbox

GreenPrint Core Practice #1: Schools practice visionary leadership.

To effectively engage the early majority in advancing climate justice and sustainability in education, it is crucial to equip district leaders with a toolbox of professional development resources and language that connects these critical issues to various roles and responsibilities across the network. This toolbox could include ready-made workshops, discussion guides, case studies, and frameworks that align sustainability with core educational goals, such as student achievement, mental health, equity, and community engagement.

Opportunity #14: Engage content experts and community members

GreenPrint Core Practice #4: Schools cultivate strong school-family-community partnerships.

The PPS Spring Climate Summits have successfully brought in content experts to deliver keynote addresses, including guests from elected offices and community-based organizations. These individuals are valuable resources for climate education. However, a knowledge gap still needs to be addressed through teacher professional

development and curriculum development. In 2024, two PPS AmeriCorps members, Cal Verstrate and Anita Spaeth, created a highly successful food justice lesson that teachers are requesting again. Additionally, several other organizations have shown interest in contributing to classroom instruction. PPS should create a directory of guest speakers and community-based organizations to support this effort further.



Opportunity #15: Gather data on student and staff mindset

GreenPrint Core Practice #10: Curriculum and instruction leaders build a comprehensive and coherent system of assessment that measures sustainability literacy and promotes health and equity.

Data around mindsets will be essential in understanding the current landscape and guiding targeted efforts. Surveys, classroom observations, walkthroughs, and analyzing the scope and sequence of curricula can provide valuable insights into educators' perceptions, beliefs, and readiness to incorporate climate justice and sustainability into their practice. Combining these strategies can create a more inclusive and widespread commitment to this work.

Opportunity #16: Continue to support and expand the School Sustainability Champions Network

GreenPrint Core Practice #2: Schools use shared leadership structures.
The district launched the SSCN in Spring 2023 to connect with and support school-

based staff (mostly teachers) with sustainability initiatives. The Network gives members a more direct line of communication to the district, and to each other. With Middle Schools and High Schools receiving PCEF Student Climate Project funds, there is an opportunity to pivot the SSCN to focus on supporting Elementary schools in more specialized ways.



Opportunity #17: Provide targeted support and gather feedback from Climate Justice and Environmental Science teachers

GreenPrint Core Practice #7: Schools create the conditions for leaders, teachers, and students to plan and implement high-quality curriculum and instruction for health, equity, and sustainability.

Teachers of Climate Justice and Environmental Science courses are currently underserved. Low enrollment and staff reductions have led to the rollback of many of these classes, which is a significant setback considering the community pressure and student advocacy that helped establish these electives. Due to limited capacity, district staff in the Office of Teaching and Learning (OTL) and Climate Justice departments have been unable to provide the level of support necessary to help these teachers make the courses successful and increase their popularity.

Resource Development and Accessibility: Insights

Insight #10: Addressing the Need for Accessible and Curated Climate Change and Sustainability Resources in Education

GreenPrint Core Practice #8: Schools use an integrated approach to curriculum design that reflects the interdisciplinary nature of the world around us

Both district leaders and classroom teachers emphasized the need for localized climate justice resources that are vetted, trustworthy, developmentally appropriate, engaging, accessible, and diverse. Some educators felt that more resources were needed, while others found the existing resources overwhelming, making it time-consuming to find suitable materials for specific grade levels or content areas. There was a clear desire for a digital space where teachers could easily share climate justice units, lessons, and projects.



Resource Development and Accessibility: Opportunities

Opportunity #18: Create a dynamic and collaborative digital resource hub to support climate justice and sustainability education

GreenPrint Core Practice #8: Schools use an integrated approach to curriculum design that reflects the interdisciplinary nature of the world around us

This hub aims to empower teachers, students, and the community by providing easy access to localized, trustworthy, and engaging resources focused on climate solutions. It will serve as a platform for sharing best practices, fostering collaboration, and highlighting success stories across the district. PPS district leaders, educators, and students will set up and maintain it. This can be the place to showcase positive outliers across the district, map out community-based partnerships, and a space to help students build knowledge across disciplines, make connections to the things they care about, and take action.

A digital resource hub would go a long way in addressing and integrating the insights and opportunities above and is a strong candidate for short- and medium-term work in implementing PPS' Climate Justice Policy. In the next few pages,, we elaborate on some of the steps involved in creating such a hub.



Possible Plan and Strategy for Four Key Leverage Points

Leverage Point #1: Digital Resource Hub

1. Crowdsourced Content Sharing Platform:

- ◊ Develop a user-friendly space for teachers to contribute lessons, units, and climate justice-related resources. This platform would allow educators to upload materials, share insights, and discuss strategies that have proven effective in their classrooms.
- ◊ Ensure the platform supports diverse media types (e.g., lesson plans, slide decks, videos, project templates) to cater to teaching and learning styles.
- ◊ Consider expanding the Portland Public Schools Climate Justice Google site to incorporate these functionalities or developing a new site with more robust features.

2. Curated Climate Justice and Sustainability Resources:

- ◊ Organize resources by grade level, topic, and geographic location to facilitate easy navigation and integration across various disciplines. This could include materials on environmental science, social studies, literature, arts, etc.
- ◊ Provide local and global context by curating resources that connect local climate justice issues to broader global movements, emphasizing how actions in Portland fit into the bigger picture.
- ◊ Integrate multimedia resources, including videos, podcasts, articles, and interactive modules, to engage different types of learners and accommodate varied educational needs.

3. Integration Across the Curriculum:

- ◊ Offer guidance and exemplars on how to weave climate justice themes into existing curricular areas, whether in STEM, humanities, arts, or physical education.
- ◊ Develop cross-disciplinary units and projects that connect subjects through a climate justice lens, enabling holistic and interconnected learning experiences.
- ◊ Include outdoor and experiential learning opportunities that link students' classroom learning with the local environment, fostering a deeper understanding of sustainability in their communities.

4. Success Story Repository and Dissemination Tool:

- ◊ Create a dedicated section to collect and disseminate success stories from across the district, celebrating innovative teaching practices, student-led initiatives, and community partnerships.
- ◊ Highlight impactful projects, particularly those funded by the Portland Clean

Energy Fund (PCEF) grants and other local initiatives, showcasing how students and schools are taking action against climate change.#

- ◊ Feature stories in various formats (e.g., blog posts, videos, infographics) to reach a wider audience and inspire others to replicate successful approaches. (Examples of teacher interviews here. Examples of students interviewing students here. Washington curates stories and resources here.)

5. Collaboration and Community Engagement:

- ◊ Facilitate collaboration among educators, students, parents, and community members by incorporating features like forums, discussion boards, and event calendars to promote ongoing dialogue and partnership.
- ◊ Leverage the hub as a communication channel for climate justice initiatives throughout Portland, fostering a sense of shared purpose and community.
- ◊ Engage local organizations, experts, and activists in contributing to the hub's content, ensuring it reflects various perspectives and expertise.

6. Implementation Strategy:

- ◊ Form a Development Team: Assemble a diverse team that includes teachers, students, district leaders, IT specialists, and community partners to guide the hub's development. Ensure the team reflects the diversity of PPS and includes voices from various grade levels, disciplines, and communities.
- ◊ Pilot and Iterate: Launch a pilot version of the hub with select schools or teacher groups, gather feedback, and iterate based on user experience. Scale up gradually, incorporating lessons learned and refining the hub's features and content.
- ◊ Promote and Sustain: Develop a communication plan to raise awareness and encourage participation among teachers, students, and community members. Consider hosting workshops, webinars, or professional development sessions to train educators on using the hub effectively.

7. Continuous Improvement and Feedback Loop:

- ◊ Establish regular updates and a content refreshment system to keep the hub dynamic and relevant. This could include teacher and student surveys, feedback forms, and periodic reviews by a dedicated team.
- ◊ Monitor usage analytics to understand the most valuable resources, continuously improving the site based on user behavior and needs.
- ◊ Encourage user feedback and create opportunities for community input to adapt and expand the hub's offerings over time.

Leverage Point #2: Implement Climate Literacy Adoption Criteria Aligned with PPS Curriculum and 7-Year Cycle

Phase 1: Define

1. Project Plan Development:

- ◊ Establish a timeline aligned with the state's 7-year adoption cycle.
- ◊ Identify key stakeholders (teachers, climate justice advocates, curriculum experts) to form the adoption committee.

2. Adoption Scope:

- ◊ Clearly define the scope of integration, ensuring climate literacy and justice education are embedded across subjects (e.g., science, social studies, English).
- ◊ Specify grade levels (Pre-K through 12) and highlight interdisciplinary opportunities for climate justice.
- ◊ Climate Literacy Position Paper:
- ◊ Develop a position paper on best practices for integrating climate literacy and justice education, referencing both state and national standards (e.g., Next Generation Science Standards).
- ◊ Include guiding principles on sustainability, social justice, equity, and climate change's impact on frontline communities.

3. Evaluation Matrix:

- ◊ Create a matrix to evaluate current curriculum materials for alignment with climate literacy and justice education.
- ◊ Include criteria for assessing coverage of key climate topics (e.g., climate science, environmental justice, sustainability practices).

Phase 2: Select

1. Initial Program Review:

- ◊ Use the evaluation matrix to review current district materials and identify gaps in climate literacy.
- ◊ Invite publishers to present their programs based on alignment with the district's climate literacy goals.

2. Narrowing Down Options:

- ◊ Select publisher programs that meet the climate literacy integration criteria.
- ◊ Conduct a "standards trace" to assess how well these programs align with the Climate Crisis Response Policy and state standards.

3. Supplemental Resource Identification:

- ◊ For areas where the current curriculum falls short, identify supplemental materials (e.g., lesson plans, modules, digital resources) to bridge gaps in climate literacy.
- ◊ Engage local and national organizations specializing in climate education to provide resources.

Phase 3: Implement

1. Professional Development:

- ◊ Develop a multi-tiered professional development plan to train teachers on integrating climate literacy and justice education into their existing curriculum.
- ◊ Offer specific training for interdisciplinary and project-based learning opportunities (e.g., outdoor education, community sustainability projects).

2. Resource Rollout:

- ◊ Ensure new instructional resources are distributed to schools along with support for teachers to integrate these into their daily lessons.
- ◊ Provide ongoing access to digital resources, climate literacy guides, and the supplementary materials identified in the Select phase.

3. On-Going Support:

- ◊ Assign dedicated instructional coaches or coordinators to provide in-classroom support for teachers implementing climate literacy.
- ◊ Offer "at-the-ready" digital tools and resources, including webinars, workshops, and model lessons.

Phase 4: Evaluate

1. Continuous Evaluation:

- ◊ Conduct regular assessments to track the effectiveness of climate literacy integration, gathering both quantitative (test scores, literacy rates) and qualitative (teacher feedback, student engagement) data.
- ◊ Use surveys and focus groups with students and teachers to evaluate the impact of climate literacy on learning outcomes and engagement.

2. Mid-Cycle Adjustments:

- ◊ Based on evaluation data, update and refine curriculum materials, professional

development plans, and supplemental resources as needed.

- ◊ Prepare for the next 7-year adoption cycle by reviewing emerging best practices and evolving climate literacy standards.

3. Long-Term Reporting:

- ◊ Align the climate literacy evaluation with the district's broader goals for sustainability and wellness, reporting progress annually to stakeholders.
- ◊ Incorporate data on how climate literacy is affecting broader district priorities, such as student health and environmental sustainability in school buildings.

Leverage Point #3: Climate Literacy and Sustainability Data Collection

Climate Literacy and Sustainability Data Collection

To support the integration of climate literacy and sustainability in Portland Public Schools (PPS), a structured approach to data collection is essential. Gathering data across multiple categories will provide valuable insights into current practices, identify areas for improvement, and foster a culture of transparency and accountability. Below are four key categories and suggested steps for data collection:

1. Sustainability Performance Dashboards

Creating Sustainability Performance Dashboards for schools and the district can make sustainability data accessible to both teachers and students. These dashboards would track metrics related to energy use, waste management, water consumption, and other ecological indicators on school grounds. This transparency would empower students to engage in solutionary and project-based learning opportunities, allowing them to set and monitor goals to reduce their school's ecological footprint. Examples of such dashboards include START by Green Schools Alliance and ArcSkoru, both of which provide user-friendly platforms for measuring and improving sustainability performance. Steps include:

- ◊ Identifying the key sustainability metrics to track (e.g., energy use, waste diversion, water conservation).
- ◊ Implementing dashboards that allow real-time access to data for students, teachers, and administrators.
- ◊ Incorporating data from these dashboards into classroom projects and school-wide sustainability goals.

2. Student Climate and Sustainability Actions and Projects

Collecting data on student climate and sustainability actions is crucial for understanding the scope and impact of student-led initiatives. This data can be gathered from classroom-based projects, green teams, or youth-led actions funded by the Portland Clean Energy Fund (PCEF). This can be accomplished using the free EcoSchools U.S. platform, hosted by the National Wildlife Federation.

Steps include:

- ◊ Developing a system for tracking and documenting student projects and actions related to climate and sustainability.
- ◊ Gathering information on the type, scale, and outcomes of these projects.
- ◊ Highlighting exemplary projects to inspire and motivate further action across the district.

3. Sustainability Beliefs and Mindsets

Understanding student and staff mindsets around climate justice and sustainability is key to fostering a broader culture of environmental responsibility. Data collection in this area could include surveys and focus groups that assess attitudes, beliefs, and levels of engagement with sustainability issues. The Green Schools National Network (GSNN) can provide guidance and examples of effective mindset surveys and tools. Steps include:

- ◊ Designing and distributing surveys to gather data on sustainability beliefs and attitudes among students, teachers, and staff.
- ◊ Analyzing the data to identify patterns, barriers, and opportunities for increasing engagement with sustainability issues.
- ◊ Using this information to tailor professional development and curriculum enhancements that address specific mindset gaps.

4. Curriculum and Instruction

Collecting data on curriculum and instruction helps map where and how climate literacy and sustainability education are already being integrated, as well as where opportunities for expansion exist. This involves identifying existing interdisciplinary connections and potential areas where sustainability can be further embedded. Steps include:

- ◊ Conducting curriculum audits to determine where sustainability and climate literacy are already taking place.
- ◊ Mapping interdisciplinary opportunities where climate justice themes can be introduced or expanded.
- ◊ Using the collected data to inform curriculum development and professional development efforts, ensuring a more comprehensive integration of climate literacy across subjects and grade levels.

Leverage Point #4: Infusing Climate Literacy and Sustainability into Existing Structures and Initiatives

Middle School Redesign

Portland Public Schools identified 18 initiatives across five themes—Integrated Adolescent Supports, Inclusive and Differentiated Teaching and Learning, Embracing Change, Racial Equity and Social Justice, and Professional Excellence and Support—aimed at transforming outcomes for middle-grade students. These initiatives are designed to support adolescent learners and improve the school experience for all 6th-8th graders. Each year, the district will prioritize and deeply support two to three of these initiatives.

SEL in Elementary School

Social and Emotional Learning (SEL) is a process that helps individuals, both young and old, acquire the knowledge, skills, and attitudes necessary to understand themselves, connect with others, achieve goals, and support their communities. The CASEL framework outlines five core competencies essential for SEL: self-awareness, self-management, social awareness, relationship skills, and responsible decision-making. These competencies help individuals manage emotions, empathize with others, build meaningful relationships, and make thoughtful choices. SEL development occurs throughout life and is influenced by supportive environments, including classrooms, families, and communities. By integrating SEL into educational and social settings, we can foster inclusive, engaging, and supportive learning spaces that prepare young people to contribute positively to society.

Other schools and districts have used SEL as a framework for integrating climate literacy in a developmentally appropriate way, especially in elementary schools. Responsible decision-making, one of the core SEL competencies, naturally aligns with the principles of doing the most good and least harm for people, the planet, and animals. By encouraging students to consider the broader impact of their choices,

teachers can weave in concepts of sustainability and environmental stewardship without directly mentioning terms like climate change or global warming. Relationship skills, another key SEL component, can be developed as students collaborate on projects aimed at improving their schools and communities, such as making them cleaner, safer, and greener. These hands-on efforts foster a sense of self-efficacy and confidence in young people, empowering them to take action. There are many academic and transdisciplinary connections that can be made with this project-based learning approach that can tie SEL together with academic skills and sustainability. By incorporating Traditional Ecological Knowledge (TEK) and biophilia—the innate human connection to nature—teachers can enhance SEL while fostering a love and respect for the environment. Since SEL is often a district-wide initiative, it serves as a powerful leverage point for embedding climate literacy into everyday learning, laying the groundwork for a more sustainable future.

CTE and other Workforce Pathways

The rise of green jobs in the U.S. is reshaping the workforce, expanding beyond traditional sectors like solar and wind energy to include industries such as construction, medicine, architecture, and even finance. As companies increasingly prioritize sustainability due to customer and shareholder demands, the need for workers with green skills has grown. In 2022, over 410,000 job postings were for "core green" roles, and it's expected that the demand for these positions will grow by 5.5% over the next five years. Beyond these core green jobs, there's a significant demand for "green-enabled" jobs in non-traditional fields that require at least one green competency. With over 51.6 million workers who could potentially be reskilled for these positions, the green economy presents enormous opportunities for workforce transformation. Upskilling programs and green job initiatives are vital to meeting the needs of this expanding sector, allowing workers to transition into roles that can drive sustainability and climate action across the economy.

Portland Public Schools (PPS) can leverage Career and Technical Education (CTE) courses and other workforce development pathways in high schools to infuse climate literacy into the curriculum in ways that connect to students' future careers and livelihoods. By reframing every job as a potential green job, educators can help students see how any interest or passion can be translated into a climate solution. This systemic integration of climate literacy can energize high school coursework and provide relevance to students' lives beyond the classroom. While some high schools already offer CTE programs related to climate and sustainability, there are

vast opportunities to scale and multiply these efforts. Expanding these examples can provide students with a broader understanding of how climate solutions can permeate all industries, preparing them for a future where green skills are increasingly necessary.

Tribal History, Shared History

Tribal History/Shared History (THSH) was established by Oregon Senate Bill 13 in 2017 to create K–12 Native American curriculum for public schools and provide professional development for educators. It also funds Oregon's nine federally recognized tribes to develop place-based curricula that reflect their unique histories and experiences. This initiative addresses the lack of accurate representation of Native American history, which has contributed to opportunity and achievement gaps for Native American students. Grounded in the Essential Understandings of Native Americans in Oregon, the curriculum aims to offer a more complete and authentic portrayal of Native American histories and perspectives, correcting longstanding misconceptions in traditional curricula.

As mentioned above in the report, the teaching of Indigenous knowledge and ways of knowing is already integral to climate literacy, as one of the core climate solutions is to protect the lands and rights of Indigenous peoples. By including Tribal History/Shared History, which is grounded in the Essential Understandings of Indigenous Lifeways, students gain a deeper understanding of the interconnectedness and reciprocity between humans and the natural world—principles central to climate literacy. Indigenous knowledge emphasizes sustainable practices, stewardship, and respect for the environment, offering critical lessons for addressing climate change. Through this curriculum, students not only learn about the history and rights of Indigenous peoples but also how their traditional ecological knowledge can inform and guide modern climate solutions.

Create a Crosswalk Between Tribal History/Shared History and the Climate Literacy Guide

1. **Form a Collaborative Working Group:** Establish a team that includes representatives from the nine federally recognized Oregon tribes, PPS curriculum specialists, climate literacy experts, and tribal liaisons. This group will oversee the development of the crosswalk, ensuring cultural sensitivity and legal compliance.
2. **Review Tribal History/Shared History Curriculum:** The working group should

conduct a comprehensive review of the Tribal History/Shared History curriculum, paying special attention to the Essential Understandings related to Indigenous Lifeways, reciprocity, and environmental stewardship.

3. **Align with Climate Literacy Principles:** Identify specific elements within the Climate Literacy Guide that align with Indigenous knowledge, particularly the principles of interconnectedness, sustainability, and the role of protecting Indigenous lands as a climate solution.

4. **Ensure Tribal Authorization for Use:** Since the lessons of Tribal History/Shared History cannot be altered without tribal authorization, the working group should engage in direct consultation with tribal governments. Any plans to integrate content from both guides must be reviewed and approved by the tribes to maintain the integrity of the material.

5. **Develop a Crosswalk Document:** Create a crosswalk that highlights areas where the two curricula naturally align, showing connections between the climate literacy competencies and Tribal History/Shared History lessons on ecology, sustainability, and human-nature relationships. This crosswalk should maintain the integrity of the tribal lessons while enhancing students' understanding of climate justice.

6. **Provide Professional Development:** Offer training sessions for educators to help them understand how to use the crosswalk, ensuring they are aware of the restrictions on altering Tribal History/Shared History content and how to integrate climate literacy without modification.

7. **Gather Feedback and Refine:** After initial implementation, gather feedback from educators, students, and tribal representatives. Use this feedback to refine the crosswalk, making adjustments while maintaining compliance with tribal authorizations.

Appendix

PPS Climate Crisis Response Policy

Below is the relevant excerpt from PPS Climate Crisis Response, Climate Justice and Sustainable Practices Policy (3.30.080-P)

Pillar 3: Provide Effective Environmental and Sustainability Education

Goal 3.1: PPS will empower staff as allies for a healthy climate.

1. Provide learning opportunities for staff on climate science/climate justice.
2. Continuously collaborate with staff to utilize sustainability practices in the performance of their work and provide information, support, and tools to help implement those practices.

Goal 3.2: With guidance from frontline students and communities, PPS will develop curricular learning opportunities, so PPS graduates know the causes and consequences of climate change, understand climate justice, and have opportunities to practice climate solutions.

1. Develop and deliver curriculum and resources to help students understand, prepare for, and respond to climate change impacts.
2. Teaching and learning about climate change will recognize and support the emotional and other impacts of climate change on individuals.
3. Provide opportunities for students to probe the underlying causes of the climate crisis and the structural racism embedded in climate change due to actions by majority white countries with disproportionate negative burdens and impacts on people of color, and to understand the dynamic nature of complex systems and how they change, and opportunities for them to practice using systems thinking to inform their choices.

Goal 3.3: PPS staff will collaborate with students to create opportunities to engage youth in hands-on climate learning, preparation, and practice on a regular basis at all PPS schools.

1. Support development of youth leadership and engagement opportunities around climate solutions, prioritizing opportunities for students from frontline

Communities.
2. Support student-led climate justice, climate action, and climate resiliency Initiatives.
3. Treat the grounds and garden of each school as a learning space and involve students in nature-based grounds improvements and design. Support opportunities for students in every PPS elementary school to learn in a designated outdoor learning space.
4. Support students in their ongoing and critical role as climate response, climate justice, and sustainability advocates.

Climate Education Ecosystem in the U.S.

The landscape of climate education in the United States is gaining momentum, reflecting an increased awareness of the need to equip young people with the knowledge and skills to understand and address climate change. Several states and cities have taken significant steps to adopt policies to embed climate education within school curricula, most commonly through science standards like the Next Generation Science Standards (NGSS). NGSS, which emphasizes climate science as part of environmental education, has been adopted by over 20 states. These standards ensure that students are introduced to the scientific principles of climate change, its causes, and its impacts on ecosystems and communities. However, the reliance on science as the primary vehicle for climate education can create a narrow lens, as climate change's broader social, economic, and justice dimensions are often sidelined.

NOAA is leading the way in defining climate literacy at the federal level through its publication, Climate Literacy: Essential Principles for Understanding and Addressing Climate Change. This guide outlines key information that individuals and communities need to understand Earth's climate, the impacts of climate change, and the solutions being implemented globally. Designed for educators, communicators, and decision-makers, it is a foundational tool for developing climate-related curricula, fostering dialogue, and informing decisions and policies.

Several U.S. states, including New Jersey, California, and Washington, have taken innovative steps to move beyond traditional science-based climate education, working to create interdisciplinary approaches that span multiple subjects. These states recognize that climate change is a multifaceted issue, requiring education incorporating scientific, social, economic, and political dimensions. Each state

has developed unique strategies to embed climate education across its curricula, empowering students to think critically about the impacts of climate change and the role they can play in addressing it.

While each of the following states has passed climate education policies, each continues implementing them with mixed results, with pockets of success and serious obstacles of funding, capacity, educational resources, and professional development. The only way to progress as a country will be to collaborate and learn from one another.

New Jersey:

In 2020, New Jersey became the first state in the U.S. to pass legislation requiring climate change to be integrated across all subjects in K-12 education. Through the New Jersey Student Learning Standards, the state seeks to ensure that students learn the science behind climate change, its local and global impacts, and how to make informed, sustainable actions. This interdisciplinary approach fosters critical thinking and action across subjects like science, social studies, and the arts. Supported by First Lady Tammy Murphy, the state has committed millions in funding to assist school districts with implementation and established the New Jersey Climate Change Education Hub to provide resources for educators. The state also created a website for Climate Change Education Resources. Another initiative that began in New Jersey but has since grown to a national and even international scope is Subject to Climate.

California:

California has taken a significant step in advancing climate education with the passage of Assembly Bill 285, signed into law by Governor Gavin Newsom on October 8, 2023. The new law requires the state's public schools to integrate the causes, effects, and solutions to climate change into the science curriculum for grades 1-6 and 7-12, with full implementation by the 2024-2025 school year. While the original bill aimed for a more interdisciplinary approach to climate education across various subjects, the passed version focuses specifically on science. However, efforts are underway, particularly through organizations like Ten Strands, to expand this mandate into other content areas and provide resources for curriculum development and climate change professional development for educators.

California's Environmental Principles and Concepts (EP&Cs), legislated under California Public Resources Code Section 71301 and California Education Code

Section 51227.3, provide a foundational framework for promoting environmental literacy by highlighting the interconnected relationship between humans and the natural world. Rather than focusing on memorization, the EP&Cs emphasize "big ideas" that inform standards-based instruction and encourage student inquiry across various disciplines. The five environmental principles and 15 supporting concepts address themes such as the dependence of human life on natural systems, the impact of human actions on ecosystems, and the continuous exchange of matter between human and natural environments. Under the legislation, the EP&Cs are integrated into content standards and curriculum frameworks for subjects like science, English language arts, social studies, and health, ensuring that students develop a comprehensive understanding of environmental issues as part of their education.

Washington:

Washington State is a national climate change education policy leader, pioneering efforts to link climate education to the Next Generation Science Standards (NGSS). In 2017, Washington and other groups, with the support of Governor Jay Inslee, secured funding for climate education through a budget proviso. This led to the creation of ClimeTime, a state-led network and grant program focused on professional development for teachers, helping them integrate climate education into their classrooms. Since its launch in 2019, ClimeTime has received over $24 million in funding, supporting educator capacity-building and the development of climate-focused instructional materials. The initiative is led by the Washington Office of Superintendent of Public Instruction (OSPI) in collaboration with the University of Washington's Institute for Science and Math Education (UW-ISME). In addition to ClimeTime, the state has invested in the Climate Teacher Education Collaborative, which supports pre-service teacher training on climate topics. Washington's climate education efforts emphasize climate justice and equity, ensuring that diverse student populations—including Tribal schools, rural areas, and underserved communities—benefit from climate-focused learning.

New York State

While New York State has and continues to struggle to pass climate education policy, the state and city have found innovative ways to lay the foundation for high-quality climate change education.

In 2023, the Climate Resilience Education Task Force (CRETF) published New York

State's first Climate Education Platform, outlining the key elements needed for a successful statewide climate education program. The platform was developed by an intergenerational committee of educators, scholars, policymakers, and students, drawing from best practices in other states like California and New Jersey and aligning with New York's ambitious Climate Leadership and Community Protection Act (CLCPA). The platform envisions a comprehensive, interdisciplinary approach to climate education in PK–12 schools, designed to support a just and equitable transition for the state's workforce, achieve climate justice, and enhance community resilience. While New Jersey is the only state to fully integrate climate education across all grade levels and content areas, New York's platform aims to connect climate education with broader state priorities, offering a unique opportunity to bridge the goals outlined in the CLCPA Scoping Plan with educational initiatives that equip students to face the climate challenges of today and tomorrow.

The New York Climate Exchange is a center for urban climate solutions located on Governors Island, launched in 2023 after a competitive selection process. As a global hub for climate action, The Exchange is crucial in advancing climate education by integrating research, workforce development, and policy initiatives to develop and implement solutions to the climate crisis. Its mission focuses on addressing urgent climate impacts and issues of environmental justice, with a strong emphasis on equity and the role of cities in climate challenges. Through collaborative partnerships with academic institutions, corporations, and non-profits, The Exchange fosters an innovative, inclusive environment that breaks down silos to rapidly develop scalable urban climate solutions, positioning itself as both a think-tank and a do-tank for climate action.

PPS might be interested in looking at The Harbor School on Governors Island for their CTE programs and how they connect them to climate change. Students engage in hands-on, career-focused learning through their CTE programs, all incorporating climate change and sustainability aspects. Freshmen are introduced to the various CTE pathways through a Harbor Class, and by their sophomore year, students choose one of eight specialized programs. These programs lead to industry certification in marine science and technology, offering a continuum of work-based learning experiences that extend beyond the classroom into real-world environments. By weaving climate change education into these programs, Harbor School prepares students not only for careers but also to address the pressing environmental challenges facing marine and coastal ecosystems.

Positive Outliers within the GSNN Network

Here are a few GSNN partners that are further along in their implementation of sustainability and climate education initiatives. While none of these schools have fully perfected their approach, each offers unique insights and valuable lessons that PPS can learn from, both in terms of their successes and challenges.

Boulder Valley Public Schools, CO

PPS might also look to Boulder Valley School District's Seal of Climate Literacy, established through Colorado Senate Bill 24-014, which provides high school students with a diploma endorsement recognizing their commitment to sustainability and climate literacy. This program connects classroom learning with real-world challenges and opportunities related to climate change, encouraging students to engage with local environmental issues through coursework and hands-on experiences. By exploring Boulder Valley's model, PPS could consider implementing a similar endorsement, allowing students to demonstrate their dedication to climate action and sustainability while gaining valuable skills and knowledge that directly benefit their communities.

In addition, PPS can look to Boulder Valley School District's five-year action plan, developed in partnership with the GSNN, as a model for guiding their sustainability and climate education efforts. Boulder Valley's plan, which sets the course for sustainability work through 2026, was created through a collaborative process involving staff interviews, community forums, focus groups, and Ad Hoc Advisory Committee feedback. The plan outlines four strategic priority areas: 1) Curriculum and Instruction, focusing on developing experiences, knowledge, skills, values, and behaviors; 2) Climate and Culture, emphasizing relationships, health, and equity; 3) Leadership, covering decision-making structures, procurement, finance, and HR systems; and 4) Operations, ensuring sustainability across all functions. Each area includes overarching and supporting goals, along with measurable indicators to track progress. By examining this model, PPS can develop a similar comprehensive plan that engages key staff, gathers meaningful data, and sets the district on a path to becoming a healthy, equitable, and sustainable learning community.

Lexington Public School Sustainability Youth Council

PPS can draw inspiration from the Bluegrass Youth Sustainability Council (BYSC) in Lexington, Kentucky. This project-based, student-led coalition empowers young leaders from ten public and private high schools to drive sustainability initiatives

across their campuses and community. Led by student facilitators and project committee chairs, BYSC fosters lasting change through partnerships with businesses, nonprofits, and elected officials at local, state, and global levels. PPS could benefit from the experiences of these student leaders by establishing a virtual partnership with BYSC, enabling their students to collaborate, share ideas, and implement similar student-driven sustainability projects in Portland, amplifying the impact of youth-led climate action in their schools and community. Lexington is

Commonground Charter Schools - New Haven, CT

At a school-site level, PPS can look to Common Ground Charter School in New Haven, CT, as a powerful example of integrating climate and sustainability education with place-based learning deeply rooted in academic excellence. Common Ground serves as a center for environmental learning and leadership, providing opportunities for students and community members to connect with their local environment through hands-on experiences. The school's unique integration of farm, forest, and city—utilizing a working farm and 20 acres of forest land—allows students to explore rural and urban ecology while developing practical paths to urban sustainability. Common Ground's curriculum fosters environmental stewardship and academic rigor, empowering students to excel in their studies while engaging with real-world challenges related to sustainability.

Core Practices of the GSNN GreenPrint™

The Leadership System

1. Schools practice visionary leadership
2. Schools use shared leadership structures
3. Schools promote a culture of community and belonging
4. Schools cultivate strong school-family-community partnerships
5. Leaders create and maintain a communication system that engages the whole community in support of healthy, equitable, and sustainable schools

The Curriculum and Instruction System

1. Schools plan strategically for curriculum and instruction that promote health, equity, and sustainability
2. Schools create the conditions for leaders, teachers, and students to plan and implement high-quality curriculum and instruction for health, equity, and sustainability
3. Schools use an integrated approach to curriculum design that reflects the

interdisciplinary nature of understanding the world around us
4. Schools use instructional practices that include phenomena-, place-, project-, and problem-based learning
5. Curriculum and instruction leaders build a comprehensive and coherent system of assessment that measures sustainability literacy and promotes health and equity

The Culture and Climate System

1. Schools plan strategically for a school culture and climate that promotes health, equity, and sustainability mindsets
2. Schools promote healthy relationships between peers, adults, the local community, and the natural world
3. Schools create a healthy, equitable, and sustainable food culture
4. Schools have programs that promote fitness, recreation, and nature exploration for students and staff
5. Schools embrace a holistic approach to wellness for teachers, staff, and students that includes physical, mental, and social-emotional health

The Facilities and Operations System

1. Schools plan strategically for building, renovating/replacing, and operating and maintaining sustainable facilities
2. Schools build leadership capacity among all stakeholders to create and operate healthy, equitable, and sustainable facilities
3. Schools design, build, and upgrade existing facilities with systems that embrace healthy, equitable, and sustainable building principles and practices
4. Schools manage and operate facilities with internal systems that reduce environmental impacts, enhance health, and promote equity
5. Schools involve the entire school community in working toward a healthy, equitable, and sustainable learning environment

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